

Quality Inspection Report for Selective Laser Melting of 316L Stainless Steel Flat Washers

Report Number: QIR-SLM-316L-2023-08-015

Date of Inspection: October 26, 2023

Inspector: J. Rodriguez, Senior Quality Engineer

Process Evaluated: Selective Laser Melting (SLM) Batch #B-2847

Product: 316L Stainless Steel Flat Washers, Lot #W-3377

1.0 Executive Summary

This report documents the quality inspection for manufacturing lot #W-3377 comprising thirty-three (33) 316L stainless steel flat washers produced via Selective Laser Melting. The inspection was conducted in accordance with internal quality procedure QP-AM-04 and references ASTM F3184-16 for additive manufacturing of metal components. All dimensional, material, and process parameters met specified requirements. The manufacturing batch demonstrated acceptable process control with proper documentation of material inputs, energy consumption, and waste streams.

Overall Finding: PASS - All inspected characteristics conform to established specifications.

2.0 Inspection Scope and Methodology

2.1 Inspection Objectives

This inspection verified: - Conformance of finished washers to dimensional specifications - Proper documentation of material consumption and process parameters - Effective management of process by-products and recycling streams - Adherence to established manufacturing procedures

2.2 Inspection Methods

- **Dimensional Verification:** Coordinate Measuring Machine (CMM) with 0.001 mm resolution
- **Visual Inspection:** 10x magnification per ASTM F3122 guidelines
- **Material Traceability:** Verification of material certificates and batch tracking
- **Process Documentation:** Review of machine logs and operator records
- **Mass Verification:** Precision balance with 0.01 g accuracy

3.0 Material and Process Verification

3.1 Raw Material Consumption

All materials used in the manufacturing process were verified against certified specifications and properly documented in the material traceability system.

Material Description	Specification	Quantity	Certification Verified
X2CrNiMo1712 stainless steel for powder atomization (316L base material)	AMS 5688	4.11 kg	Yes - Cert #M-22841
Process water for water atomization	ASTM D1193 Type IV	16.8 kg	Yes - Lab Report #W-7742

Note: Water quality parameters (conductivity, TOC) were within specification limits per internal testing.

3.2 Process Gases

Shielding and processing gases were supplied per established procedures with continuous monitoring of purity levels.

Gas Type	Purity Specification	Quantity	Delivery System
Argon shielding and processing gas	99.998% minimum	3.08 kg	Bulk supply with in-line purification

Process Context: The argon consumption reflects standard operating parameters including chamber purging (700 L initial fill) and continuous flow during processing (54 L per component). Gas purity was maintained above 99.998% throughout the build cycle.

3.3 Energy Consumption

Electrical energy consumption was monitored and recorded through the facility’s power monitoring system.

Process Stage	Energy Consumption	Equipment	Monitoring Method
Water atomization of 316L powder	2.55 kWh	Atomization system #A-7	Direct meter reading
Selective Laser Melting process	64.92 kWh	SLM 280HL System #AM-12	Integrated power monitor

Historical Comparison: Previous similar batch (Lot #W-3312, June 2023) showed SLM energy consumption of 67.4 kWh, representing a 3.7% improvement in energy efficiency for the current lot.

Process Context: The SLM system operated for 13.38 hours with nominal power consumption of 5.5 kW during laser operation and 3.5 kW during non-processing periods. The water atomization energy intensity was calculated at 2.23 MJ/kg of powder produced.

4.0 Product Quality Assessment

4.1 Finished Product Specifications

Characteristic	Specification	Actual (Average)	Tolerance	Conformance
Total mass of finished washers	0.61 kg	0.61 kg	±0.01 kg	PASS
Quantity produced	33 units	33 units	Exact count	PASS
Outer diameter	20.0 mm	20.01 mm	±0.05 mm	PASS
Inner diameter	10.5 mm	10.49 mm	±0.05 mm	PASS
Thickness	2.0 mm	2.01 mm	±0.03 mm	PASS
Surface roughness (Ra)	≤10 µm	8.2 µm	-	PASS

4.2 Dimensional Verification Results

All thirty-three washers underwent 100% dimensional inspection using CMM. The results demonstrated excellent dimensional stability and repeatability across the entire build plate.

Sample Measurements (5 of 33 washers):

Washer ID	Outer Diameter (mm)	Inner Diameter (mm)	Thickness (mm)	Flatness (mm)
W-3377-01	20.012	10.492	2.008	0.008
W-3377-08	20.009	10.488	2.013	0.006
W-3377-15	20.015	10.491	2.011	0.009
W-3377-22	20.008	10.493	2.009	0.007
W-3377-29	20.011	10.489	2.012	0.005

All measurements within specified tolerances. Maximum deviation from nominal: 0.015 mm.

4.3 Visual and Surface Inspection

Visual examination under 10x magnification revealed consistent surface finish with no evidence of: - Unfused powder particles - Surface cracking or delamination - Significant porosity - Discoloration or oxidation

The as-built surface exhibited characteristic SLM texture with uniform bead appearance. No post-processing (e.g., machining, blasting) was performed prior to inspection.

5.0 By-Product Management and Material Recovery

5.1 Material Recovery Streams

The manufacturing process incorporates established procedures for material recovery and reuse to optimize resource utilization.

Material Stream	Quantity	Handling Procedure	Destination
316L powder reused within SLM process	2.94 kg	Sieved and blended with virgin powder	Returned to SLM powder feed system
316L powder returned for remelting	0.15 kg	Collected in dedicated containers	Sent to water atomization facility
Recovered process water from water atomization	16.4 kg	Filtration and treatment	Returned to process water system

Process Note: The powder reuse rate of approximately 71.5% (2.94 kg of 4.11 kg total input) aligns with established powder management protocols. All recovered powder undergoes sieve analysis and chemical verification before reuse.

5.2 Waste Disposal

Non-recoverable materials were properly characterized and disposed of according to environmental regulations.

Waste Stream	Quantity	Disposition Method	Documentation
Solid waste from water atomization	0.41 kg	Landfill disposal	Waste manifest #WM-44892
Non-recyclable 316L powder from SLM	0.01 kg	Landfill disposal	Waste manifest #WM-44893

Compliance Note: All waste streams were characterized as non-hazardous per TCLP testing. Disposal was conducted through approved waste management vendors with complete chain of custody documentation.

6.0 Process Parameter Verification

6.1 SLM Process Parameters

Critical process parameters were monitored and recorded throughout the build cycle. All parameters remained within established control limits.

Parameter	Setpoint	Actual Range	Control Limit	Status
Laser power	275 W	273-277 W	±10 W	Within limits
Scan speed	800 mm/s	795-805 mm/s	±15 mm/s	Within limits
Layer thickness	30 µm	30 µm	Fixed	Within limits
Build chamber oxygen	<0.1%	0.03-0.07%	<0.1%	Within limits
Build plate temperature	80°C	79-81°C	±3°C	Within limits

6.2 Build Chamber Environment

The argon atmosphere was maintained throughout the process with the following characteristics: - Initial chamber purge: 700 liters - Continuous flow rate: Equivalent to 54 liters per component - Oxygen content: Consistently below 0.1% - Pressure: Maintained at 10-15 mbar above ambient

The total argon consumption of 3.08 kg reflects standard operation for a build of this duration and component count.

7.0 Non-Conformities and Corrective Actions

No major non-conformities were identified during this inspection. One minor observation was noted:

Observation #1: Powder handling documentation showed a 15-minute gap in humidity recording during transfer operations.

Action Taken: Procedure QP-PH-02 was reviewed with the responsible technician. Additional training on continuous monitoring requirements was completed. No product quality impact was identified.

8.0 Conclusions and Compliance Statement

Based on the comprehensive inspection conducted, the following conclusions are drawn:

- All thirty-three (33) 316L stainless steel flat washers conform to specified dimensional, visual, and material requirements.
- Manufacturing process parameters were maintained within established control limits throughout the production cycle.

3. Material consumption, energy usage, and by-product management were properly documented and aligned with standard operating procedures.
4. The process demonstrated effective material utilization with 71.5% powder reuse and minimal waste generation.
5. All quality records are complete and properly maintained.

Compliance Statement: This manufacturing lot meets all requirements specified in drawing #D-316L-WASHER-20x10.5x2.0 and associated quality plan QP-AM-316L-01. The components are approved for shipment to the customer.

Approvals:

Role	Name	Signature	Date
Quality Engineer	J. Rodriguez	<i>Electronic Signature</i>	10/26/2023
Production Supervisor	M. Chen	<i>Electronic Signature</i>	10/26/2023
Quality Manager	R. Johnson	<i>Electronic Signature</i>	10/27/2023

Distribution: Quality Department, Production Records, Customer File (as required)

This report represents the findings at the time of inspection. Retain for a minimum of 10 years per quality records retention policy QP-RET-03.